

ORION 1.0

Problem Statement ID: ORION-PS-03

Problem Statement Title: SylvaSense

Team ID: [FILL THIS IN]

Team Name: Order of One



Proposed Solution

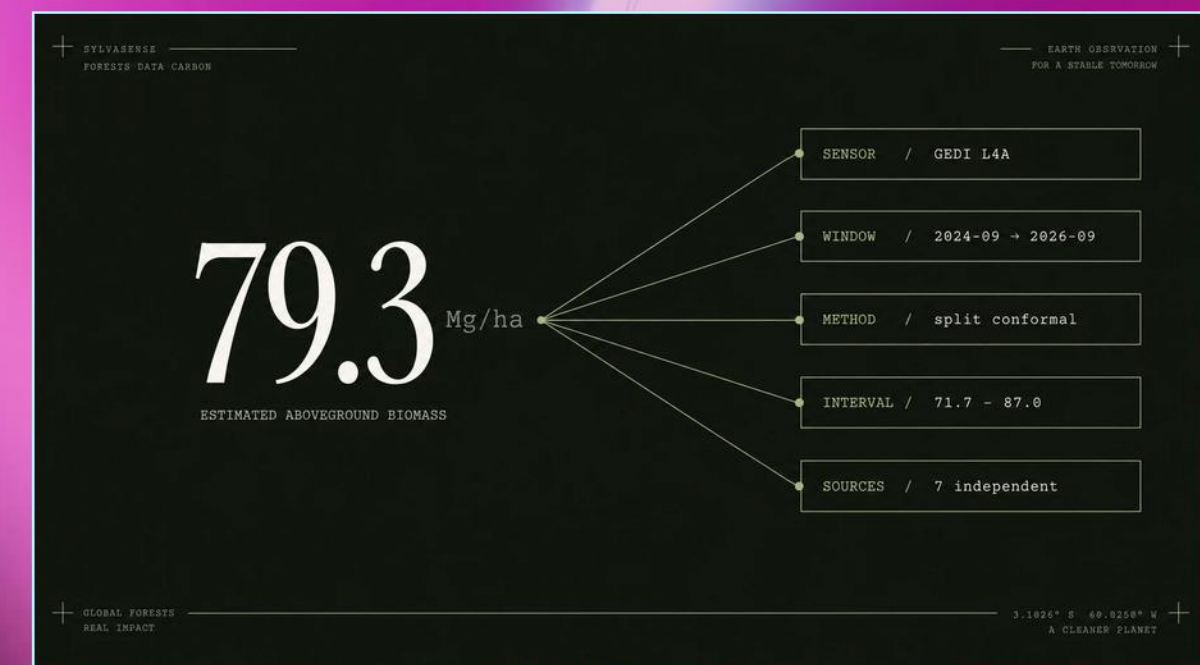
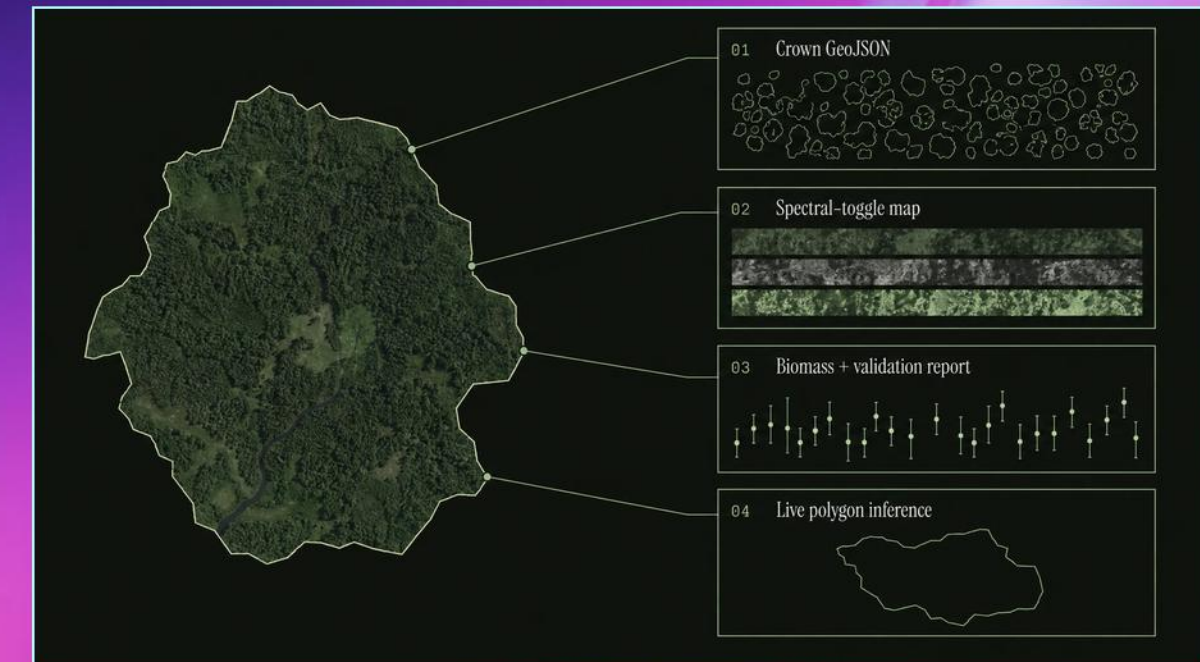
SYLVASENSE

From canopy to carbon.

One polygon in. Four kinds of evidence out.
Every number carries its uncertainty.

- 01 Crown GeoJSON**
Per-tree polygons, each with a confidence score
- 02 Spectral-toggle map**
Optical, radar and structure in one frame
- 03 Biomass + validation report**
Mg/ha with an interval that is checked
- 04 Live polygon inference**
Draw an area; evidence returns with its sources

Live demo → sylvasense-orion.vercel.app



Four outputs from one request • Crown-scale aerial imagery + regional satellite monitoring • Live system, first results on 6 sites



Technical Approach

Three scales. One pipeline.

CROWNS / FINE AERIAL IMAGERY

Mask2Former + Swin-S

0.1 m NEON RGB, tiled and merged

Every mask checked against the canopy height model

BIOMASS / OPTICAL + SAR + LIDAR

Gradient boosting + split conformal

GEDI footprints are the calibration target

Coverage measured on held-out ground, not assumed

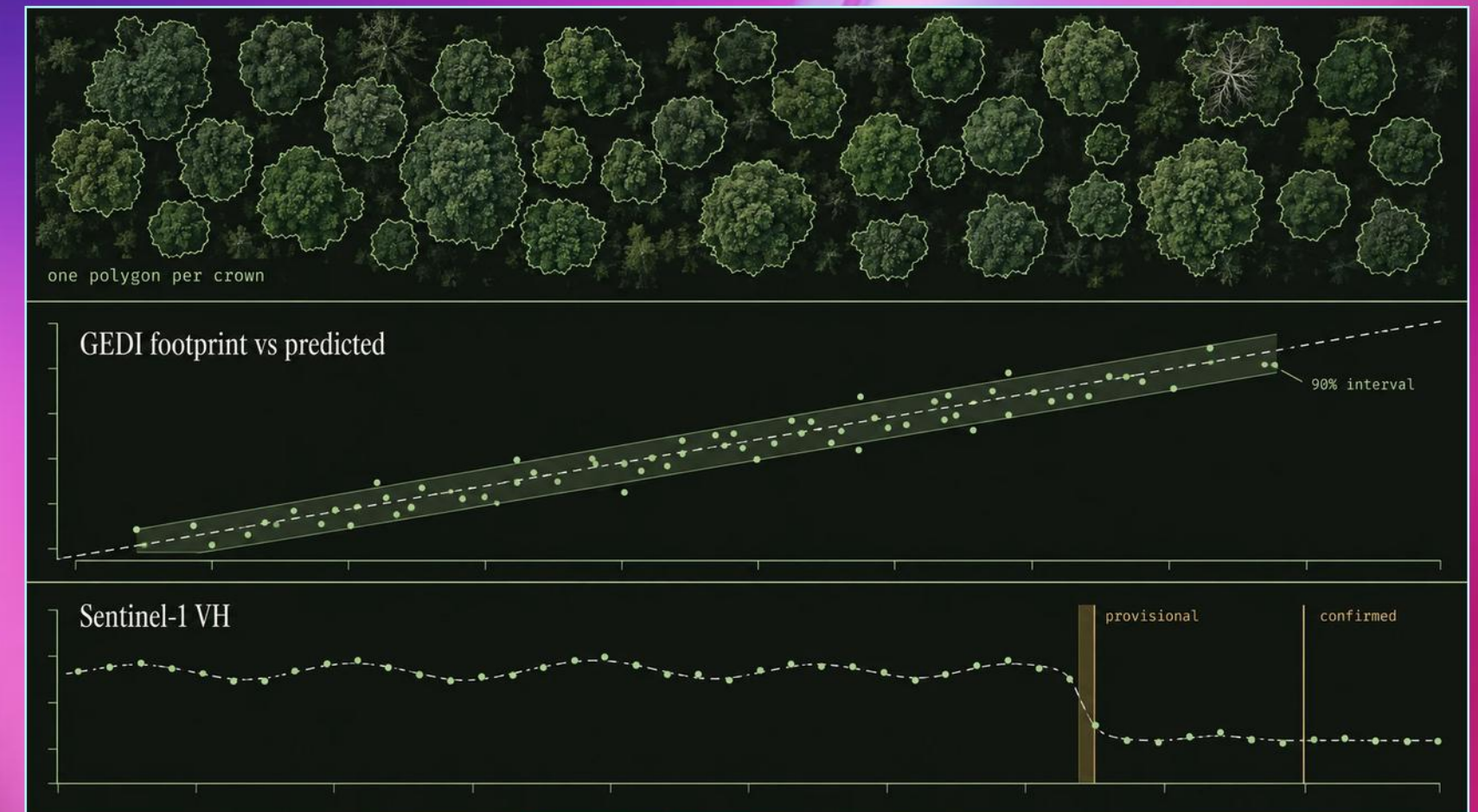
CHANGE / MULTI-YEAR OBSERVATIONS

Harmonic baseline + Bayesian confirmation

A drop must persist before it is called a detection

Below its power threshold the detector reports nothing

AGB = Σ density $\times$ area | **Aboveground carbon = AGB $\times$ CF**





Feasibility & Viability

Three gates. Already passing.

01 VERIFIED THE DATA

6 sites audited on 3 independent layers

2 of 5 candidates failed the audit and were rejected

02 EARNED THE MODEL CHOICE

Spatially-blocked folds, never a random split

Neighbouring footprints are one observation, not two

03 MEASURED THE TRADE-OFFS

False positives on primary forest: 0.01–0.02%

Coverage holds on unseen blocks — and one site fails, visibly

41.6% → 0.01%

A 365-day window was confirming a national park as cleared.
Persistence is a question about time. The gate is now a span.

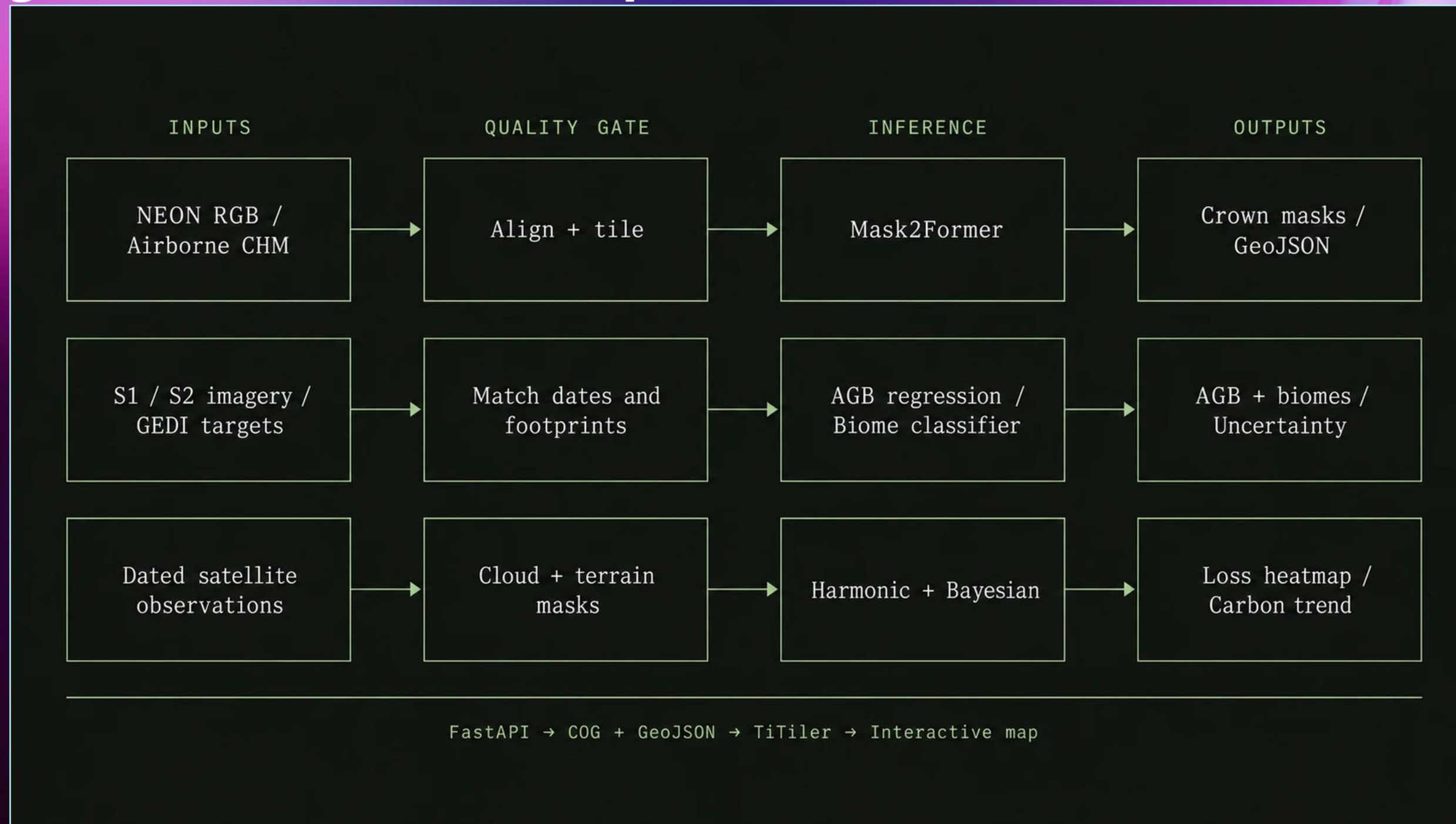
Measured on 6 sites, 2024–2026 • 60 automated tests • Rondônia, Brazil





Architecture Diagram

One polygon. Three coordinated paths.





Tech stack

DATA

Earth Engine · Planetary Computer · GEDI

PREPARE

Rasterio · GDAL · GeoPandas · odc-stac

MODEL

PyTorch · Mask2Former · split conformal

SERVE

FastAPI · Redis / Celery · Docker

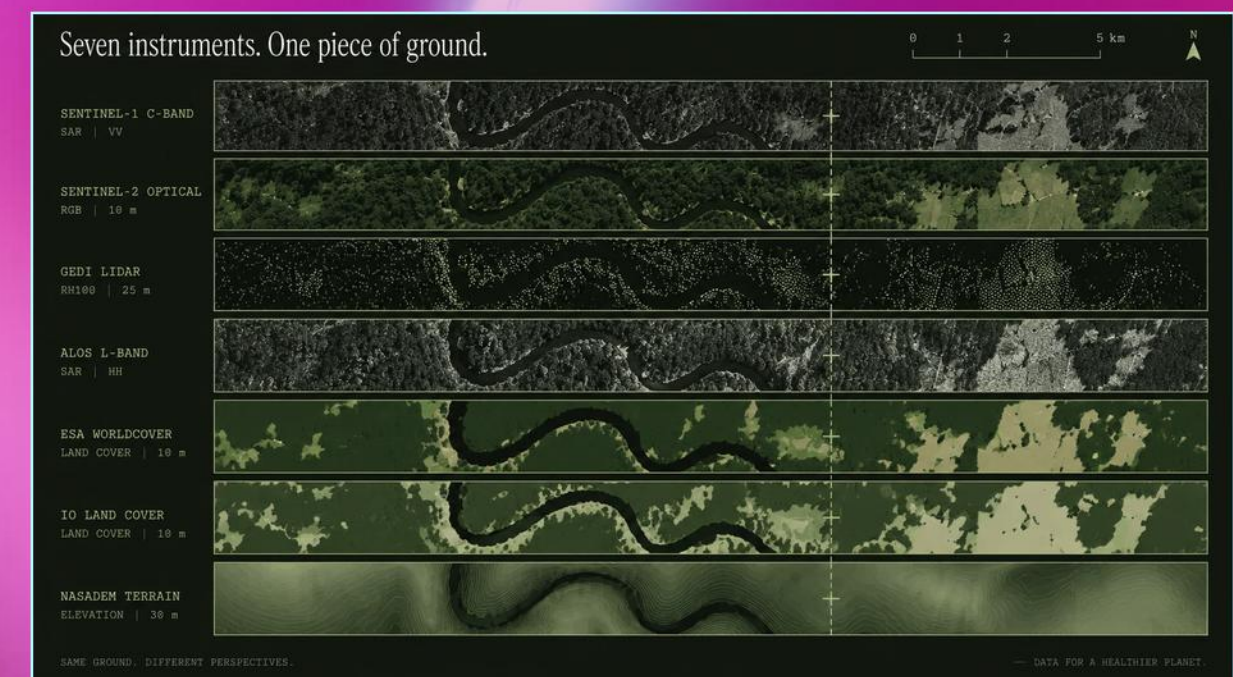
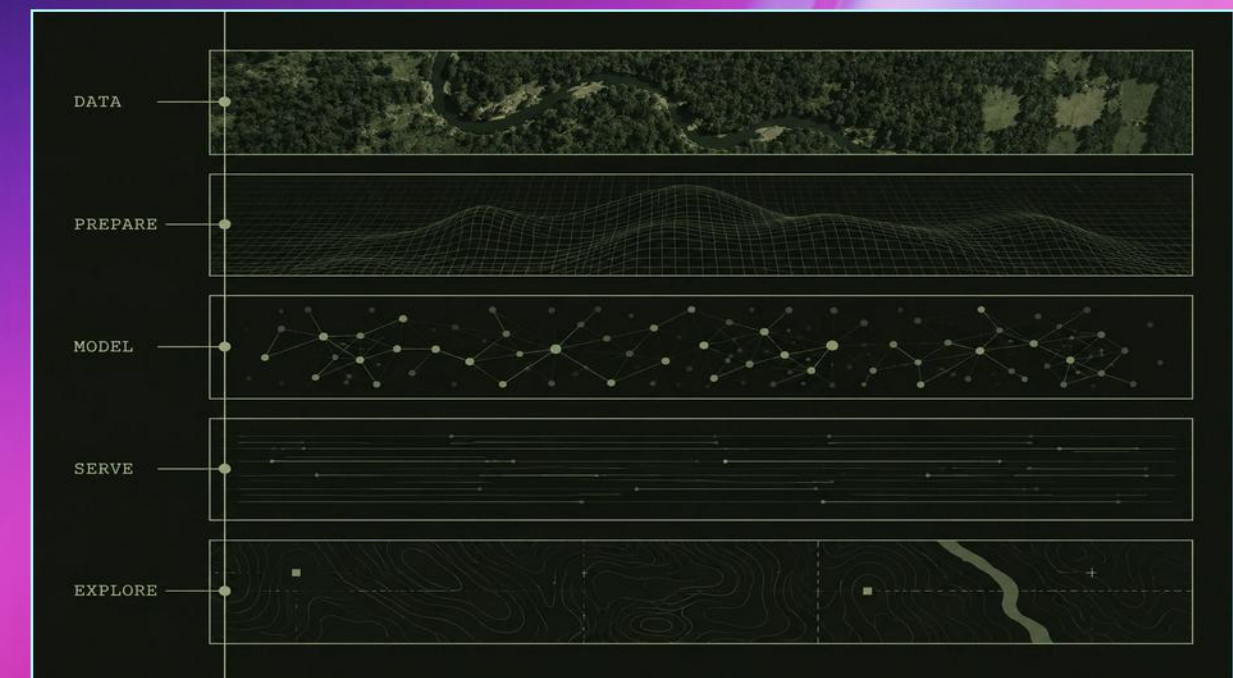
EXPLORE

TiTiler · Mapbox GL JS · deck.gl

Windowed COG reads. Cached reports.

Cold report 151–222 s · cached 3–23 ms

7 sources, each free to fail alone



Prescribed stack, with task queues and COG tile serving added for practical inference [1,7,8]

Impact & Benefits

Evidence for every forest decision.

FOREST TEAMS

Prioritise persistent loss alerts

Confirmed and provisional kept apart

RESTORATION TEAMS

Track canopy and biomass trends

Against a measured 2020 baseline

CARBON ANALYSTS

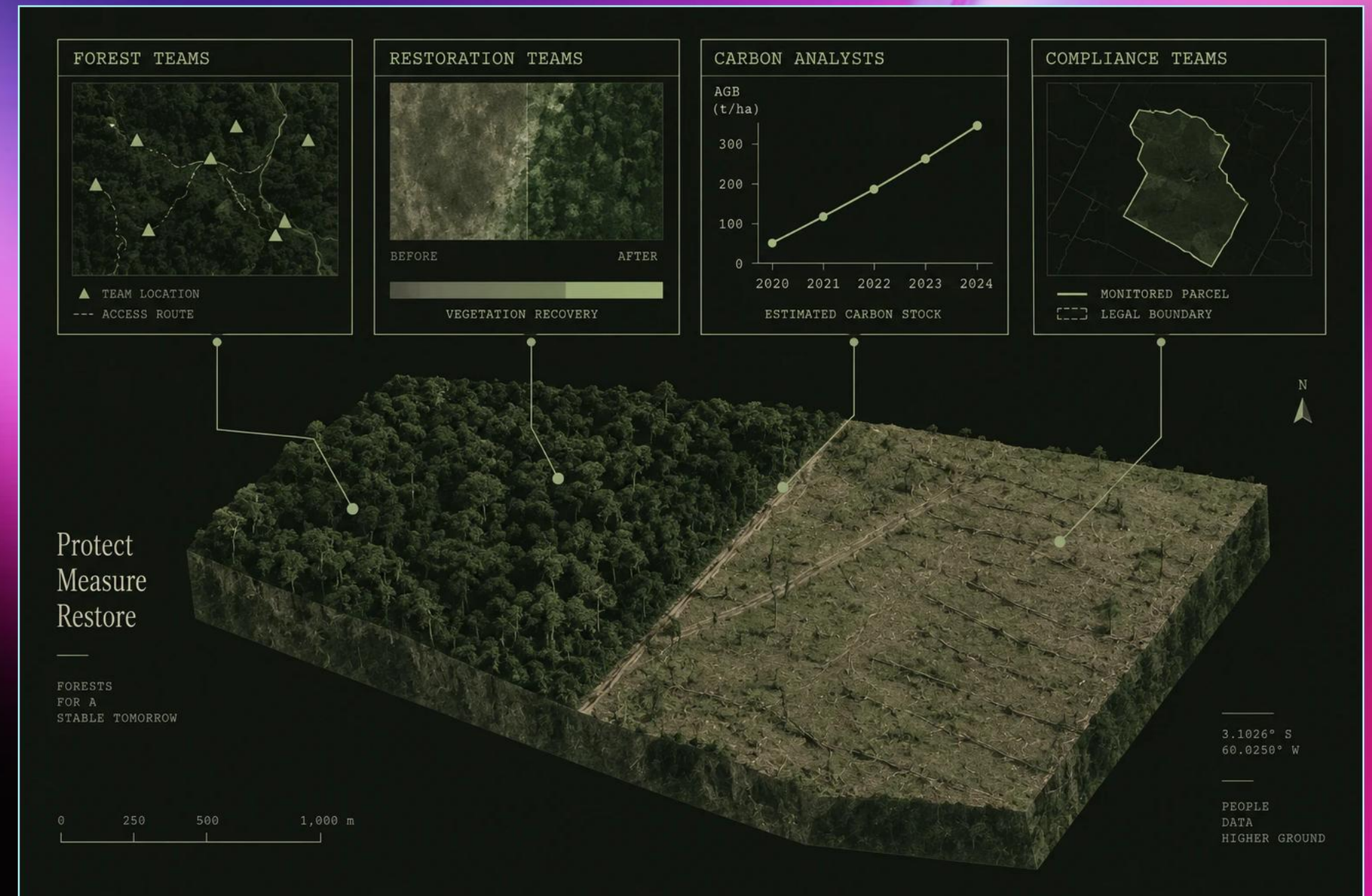
Trace stock change with its interval

Bias-adjusted area, not pixel counts

COMPLIANCE TEAMS

Evidence for EUDR, due Dec 2026

ESA WorldCover 2020 is the cut-off layer





Research & References

01 Official scope + submission rules

ORION PS-03 · Microsoft Club SIST

02 Satellite observations

Sentinel-1 processing

Sentinel-2 resolution

03 Biomass reference targets

NASA GEDI L4A · footprints + quality

04 Crown-scale data

NEON 0.1 m RGB

NEON 1 m CHM

05 Mask2Former + crown evidence

Cheng et al., 2022

Nakada et al., 2026 preprint

06 Crown segmentation benchmark

Detectree2 · Ball et al., 2023

07 Quantile biomass regression

XGBoost · official quantile objective

08 Time series + raster serving

Earth Engine CCDC

TiTiler

09 Biome labels

RESOLVE ecoregions · Dinerstein et al., 2017

10 Biomass-to-carbon formulation

IPCC 2006 · Forest Land · Table 4.3